

Assignment No 2

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Subject :- Essentials of Data Science (EDS)

Div : SE1 Batch : S15

TITLE : FIFA Player Performance Analysis Dataset

PROBLEM STATEMENT :

Analyze the performance of top football players using key metrics such as goals, assists, matches played, age, and ratings. The objective is to:

- Identify **top-performing players** based on goals and overall rating
- Compare **player efficiency** using metrics like goals per match
- Understand how **age impacts performance**
- Perform **statistical and numerical analysis** using Pandas and NumPy
- Extract meaningful insights to support **data-driven decisions in sports analytics**

PANDAS OPERATIONS

Original Dataset

```
index.py X
index.py > ...
1 import pandas as pd
2
3 # -----
4 # CREATE FIFA DATASET
5 # -----
6 data = {
7     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
8               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
9     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
10    "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
11    "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
12    "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
13    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
14 }
15
16 df = pd.DataFrame(data)
17
18 print("\n===== FIFA DATASET =====")
19 print(df)
20
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
===== FIFA DATASET =====
   Player  Age  Goals  Assists  Matches  Rating
0   Messi   36     30      20      35     9.1
1  Ronaldo   38     28      10      34     8.9
2   Neymar   32     20      15      30     8.7
3   Mbappe   25     35      12      32     9.2
4  Haaland   24     40       8      33     9.3
5   Modric   38     10      18      36     8.5
6   Salah   32     25      14      34     8.8
7    Kane   30     27       9      35     8.6
8 De Bruyne  33     15      22      36     9.0
9 Lewandowski 35     33       7      34     8.9
```

1. HEAD (First 5 rows)

```
index.py X
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n1. HEAD (First 5 rows)")
16 print(df.head())
17
18
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

1. HEAD (First 5 rows)
   Player  Age  Goals  Assists  Matches  Rating
0   Messi   36     30      20      35     9.1
1  Ronaldo   38     28      10      34     8.9
2   Neymar   32     20      15      30     8.7
3   Mbappe   25     35      12      32     9.2
4  Haaland   24     40       8      33     9.3
PS C:\Users\x\Desktop\Operations FIFA>
```

2. DESCRIBE (Statistical Summary)

```
index.py X
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n2. DESCRIBE (Statistical Summary)")
16 print(df.describe())
17
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

2. DESCRIBE (Statistical Summary)

	Age	Goals	Assists	Matches	Rating
count	10.000000	10.000000	10.000000	10.000000	10.000000
mean	32.300000	26.300000	13.500000	33.900000	8.900000
std	4.877385	9.18997	5.212165	1.852926	0.258199
min	24.000000	10.000000	7.000000	30.000000	8.500000
25%	30.500000	21.250000	9.250000	33.250000	8.725000
50%	32.500000	27.500000	13.000000	34.000000	8.900000
75%	35.750000	32.250000	17.250000	35.000000	9.075000
max	38.000000	40.000000	22.000000	36.000000	9.300000

PS C:\Users\x\Desktop\Operations FIFA>

3. SORT BY GOALS

```
index.py X
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n3. SORT BY GOALS")
16 print(df.sort_values(by="Goals", ascending=False))
17
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

3. SORT BY GOALS

	Player	Age	Goals	Assists	Matches	Rating
4	Haaland	24	40	8	33	9.3
3	Mbappe	25	35	12	32	9.2
9	Lewandowski	35	33	7	34	8.9
0	Messi	36	30	20	35	9.1
1	Ronaldo	38	28	10	34	8.9
7	Kane	30	27	9	35	8.6
6	Salah	32	25	14	34	8.8
2	Neymar	32	20	15	30	8.7
8	De Bruyne	33	15	22	36	9.0
5	Modric	38	10	18	36	8.5

PS C:\Users\x\Desktop\Operations FIFA>

4. FILTER PLAYERS WITH GOALS > 25

```
index.py x
index.py > ...
1 import pandas as pd
2
3
4 data = {
5     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
6               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
7     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
8     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
9     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
10    "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
11    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
12 }
13
14 df = pd.DataFrame(data)
15
16 print("\n4. FILTER PLAYERS WITH GOALS > 25")
17 print(df[df["Goals"] > 25])
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

4. FILTER PLAYERS WITH GOALS > 25

	Player	Age	Goals	Assists	Matches	Rating
0	Messi	36	30	20	35	9.1
1	Ronaldo	38	28	10	34	8.9
3	Mbappe	25	35	12	32	9.2
4	Haaland	24	40	8	33	9.3
7	Kane	30	27	9	35	8.6
9	Lewandowski	35	33	7	34	8.9

PS C:\Users\x\Desktop\Operations FIFA>

5. ADD NEW COLUMN (Goal per Match)

```
index.py x
index.py > ...
1 import pandas as pd
2
3
4 data = {
5     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
6               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
7     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
8     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
9     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
10    "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
11    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
12 }
13
14 df = pd.DataFrame(data)
15
16 print("\n5. ADD NEW COLUMN (Goal per Match)")
17 df["Goal_per_Match"] = df["Goals"] / df["Matches"]
18 print(df)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

5. ADD NEW COLUMN (Goal per Match)

	Player	Age	Goals	Assists	Matches	Rating	Goal_per_Match
0	Messi	36	30	20	35	9.1	0.857143
1	Ronaldo	38	28	10	34	8.9	0.823529
2	Neymar	32	20	15	30	8.7	0.666667
3	Mbappe	25	35	12	32	9.2	1.093750
4	Haaland	24	40	8	33	9.3	1.212121
5	Modric	38	10	18	36	8.5	0.277778
6	Salah	32	25	14	34	8.8	0.735294
7	Kane	30	27	9	35	8.6	0.771429
8	De Bruyne	33	15	22	36	9.0	0.416667
9	Lewandowski	35	33	7	34	8.9	0.970588

PS C:\Users\x\Desktop\Operations FIFA>

6. GROUP BY AGE (Average Goals)

```
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n6. GROUP BY AGE (Average Goals)")
16 print(df.groupby("Age")["Goals"].mean())
```

6. GROUP BY AGE (Average Goals)

Age	Goals
24	40.0
25	35.0
30	27.0
32	22.5
33	15.0
35	33.0
36	30.0
38	19.0

Name: Goals, dtype: float64
PS C:\Users\x\Desktop\Operations FIFA>

7. FIND MAX GOALS PLAYER

```
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n7. FIND MAX GOALS PLAYER")
16 print(df.loc[df["Goals"].idxmax()])
```

7. FIND MAX GOALS PLAYER

Player	Age	Goals	Assists	Matches	Rating
Haaland	24	40	8	33	9.3

Name: 4, dtype: object
PS C:\Users\x\Desktop\Operations FIFA>

8. SELECT SPECIFIC COLUMNS

```
index.py X
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n8. SELECT SPECIFIC COLUMNS")
16 print(df[["Player", "Goals", "Assists"]])
17
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
8. SELECT SPECIFIC COLUMNS
   Player  Goals  Assists
0    Messi     30      20
1  Ronaldo     28      10
2   Neymar     20      15
3   Mbappe     35      12
4  Haaland     40       8
5   Modric     10      18
6   Salah     25      14
7    Kane     27       9
8 De Bruyne     15      22
9 Lewandowski    33       7
PS C:\Users\x\Desktop\Operations FIFA>
```

9. RENAME COLUMN

```
index.py X
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n9. RENAME COLUMN")
16 df_renamed = df.rename(columns={"Goals": "Total_Goals"})
17 print(df_renamed.head())
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

9. RENAME COLUMN
   Player  Age  Total_Goals  Assists  Matches  Rating
0    Messi   36         30      20      35     9.1
1  Ronaldo   38         28      10      34     8.9
2   Neymar   32         20      15      30     8.7
3   Mbappe   25         35      12      32     9.2
4  Haaland   24         40       8      33     9.3
PS C:\Users\x\Desktop\Operations FIFA>
```

10. DROP COLUMN (Assists)

```
index.py > ...
1 import pandas as pd
2
3 data = {
4     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
5               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
6     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
7     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
8     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
9     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
10    "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
11 }
12
13 df = pd.DataFrame(data)
14
15 print("\n10. DROP COLUMN (Assists)")
16 print(df.drop(columns=["Assists"]))
17
```

10. DROP COLUMN (Assists)

	Player	Age	Goals	Matches	Rating
0	Messi	36	30	35	9.1
1	Ronaldo	38	28	34	8.9
2	Neymar	32	20	30	8.7
3	Mbappe	25	35	32	9.2
4	Haaland	24	40	33	9.3
5	Modric	38	10	36	8.5
6	Salah	32	25	34	8.8
7	Kane	30	27	35	8.6
8	De Bruyne	33	15	36	9.0
9	Lewandowski	35	33	34	8.9

PS C:\Users\x\Desktop\Operations FIFA>

NUMPY OPERATIONS

1. Mean

```
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n1. MEAN OF GOALS")
19 print(np.mean(goals))
20
21
```

1. MEAN OF GOALS

26.3

PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

2. Median

```
index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n2. MEDIAN OF GOALS")
19 print(np.median(goals))
20
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

2. MEDIAN OF GOALS
27.5
PS C:\Users\x\Desktop\Operations FIFA>
```

3. Standard Deviation

```
index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n3. STANDARD DEVIATION OF GOALS")
19 print(np.std(goals))
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

3. STANDARD DEVIATION OF GOALS
8.718371407550839
PS C:\Users\x\Desktop\Operations FIFA>
```


4. Maximum Value

```
index.py X
index.py > ...
4     "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n4. MAX GOALS")
19 print(np.max(goals))
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

4. MAX GOALS
40
PS C:\Users\x\Desktop\Operations FIFA>

5. Minimum Value

```
index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n5. MIN GOALS")
19 print(np.min(goals))
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

5. MIN GOALS
10
PS C:\Users\x\Desktop\Operations FIFA>

6. Element-wise Addition

```

index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n6. GOALS + ASSISTS")
19 print(goals + assists)
20
21
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

6. GOALS + ASSISTS
[50 38 35 47 48 28 39 36 37 40]
PS C:\Users\x\Desktop\Operations FIFA>

```

7. Scalar Multiplication

```

index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n7. GOALS * 2")
19 print(goals * 2)
20
21

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

```

7. GOALS * 2
[60 56 40 70 80 20 50 54 30 66]
PS C:\Users\x\Desktop\Operations FIFA>

```

8. PLAYERS WITH AGE > 30

```
index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n8. PLAYERS WITH AGE > 30")
19 print(ages > 30)
20
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

8. PLAYERS WITH AGE > 30
[ True True True False False True True False True True]
PS C:\Users\x\Desktop\Operations FIFA>
```

9. Normalization (Min-Max Scaling)

```
index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n9. NORMALIZED GOALS")
19 normalized_goals = (goals - np.min(goals)) / (np.max(goals) - np.min(goals))
20 print(normalized_goals)

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

9. NORMALIZED GOALS
[0.66666667 0.6         0.33333333 0.83333333 1.         0.
 0.5         0.56666667 0.16666667 0.76666667]
PS C:\Users\x\Desktop\Operations FIFA>
```

10. Dot Product

```
index.py X
index.py > ...
1 import numpy as np
2 data = {
3     "Player": ["Messi", "Ronaldo", "Neymar", "Mbappe", "Haaland",
4               "Modric", "Salah", "Kane", "De Bruyne", "Lewandowski"],
5     "Age": [36, 38, 32, 25, 24, 38, 32, 30, 33, 35],
6     "Goals": [30, 28, 20, 35, 40, 10, 25, 27, 15, 33],
7     "Assists": [20, 10, 15, 12, 8, 18, 14, 9, 22, 7],
8     "Matches": [35, 34, 30, 32, 33, 36, 34, 35, 36, 34],
9     "Rating": [9.1, 8.9, 8.7, 9.2, 9.3, 8.5, 8.8, 8.6, 9.0, 8.9]
10 }
11 # Convert to NumPy arrays
12 ages = np.array(data["Age"])
13 goals = np.array(data["Goals"])
14 assists = np.array(data["Assists"])
15 matches = np.array(data["Matches"])
16 ratings = np.array(data["Rating"])
17
18 print("\n10. DOT PRODUCT (GOALS & ASSISTS)")
19 print(np.dot(goals, assists))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\x\Desktop\Operations FIFA> python -u "c:\Users\x\Desktop\Operations FIFA\index.py"

10. DOT PRODUCT (GOALS & ASSISTS)
3254
PS C:\Users\x\Desktop\Operations FIFA>